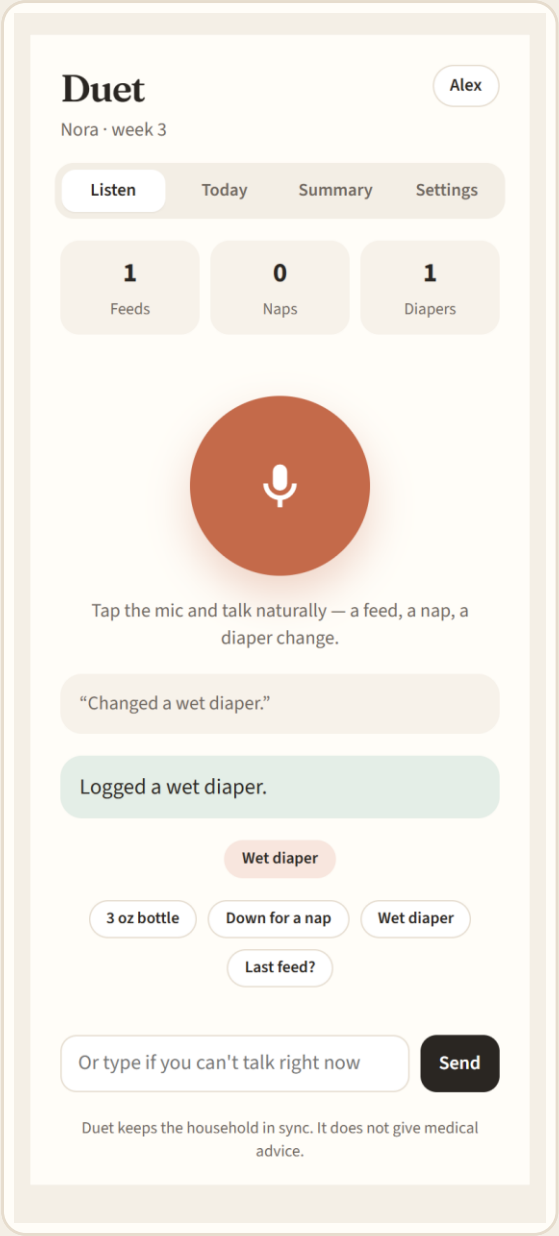
Duet

A voice-first baby log for the whole household

Talk. Log. Stay in sync.

AI Integration Class · Project presentation



THE PROBLEM

The first 12 weeks are a blur

Hands are full

Feeding, rocking, burping — there is rarely a free hand to tap through a form at 3 AM.

Care is shared

Parents, grandparents and night nurses each hold part of the picture. "Did she eat yet?" lives in group texts.

Trackers are form-heavy

Popular apps are built around charts and dashboards. Logging one bottle takes several taps.

Goal **Logging should take one sentence, and everyone should see it instantly.**

THE SOLUTION

Talk. Log. Stay in sync.

- **Speak naturally** "Just gave her a 3 ounce bottle" becomes a logged feed.
- **Everyone sees it live** Every caregiver's screen updates within a second.
- **Ask the log** "When did she last eat?" is answered from today's entries.
- **Never gives medical advice** Symptom or medicine questions are redirected to the pediatrician.
- **Free** No paywall, no credit card. Invite as many caregivers as you need.

Example

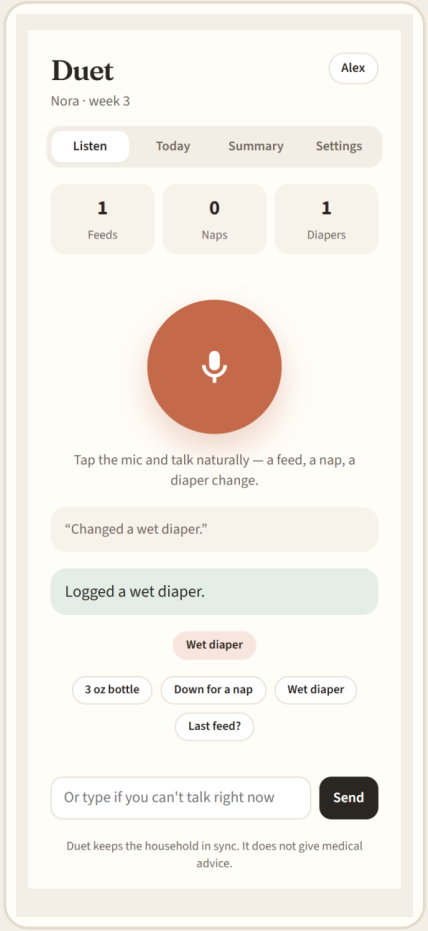
You say: "Just gave her a 3 ounce bottle."

Duet: "Logged bottle, 3 oz."

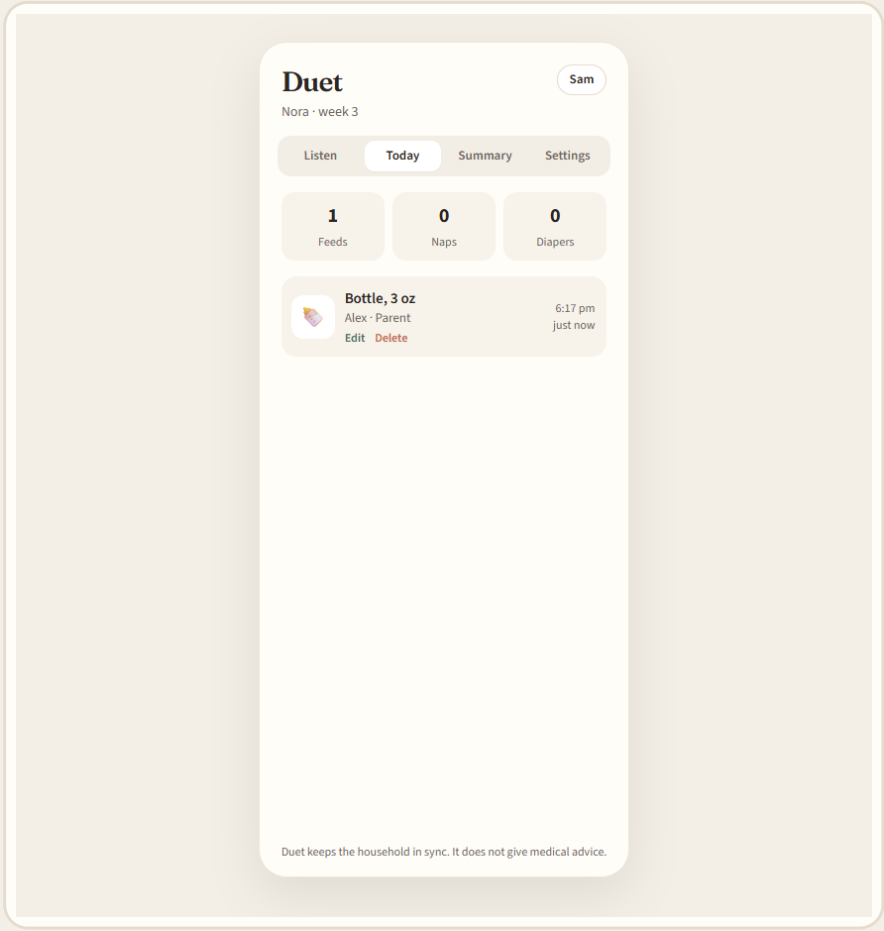
Partner's phone updates instantly: "Bottle, 3 oz — Alex · just now"

WHAT IT LOOKS LIKE

One household, two screens, live



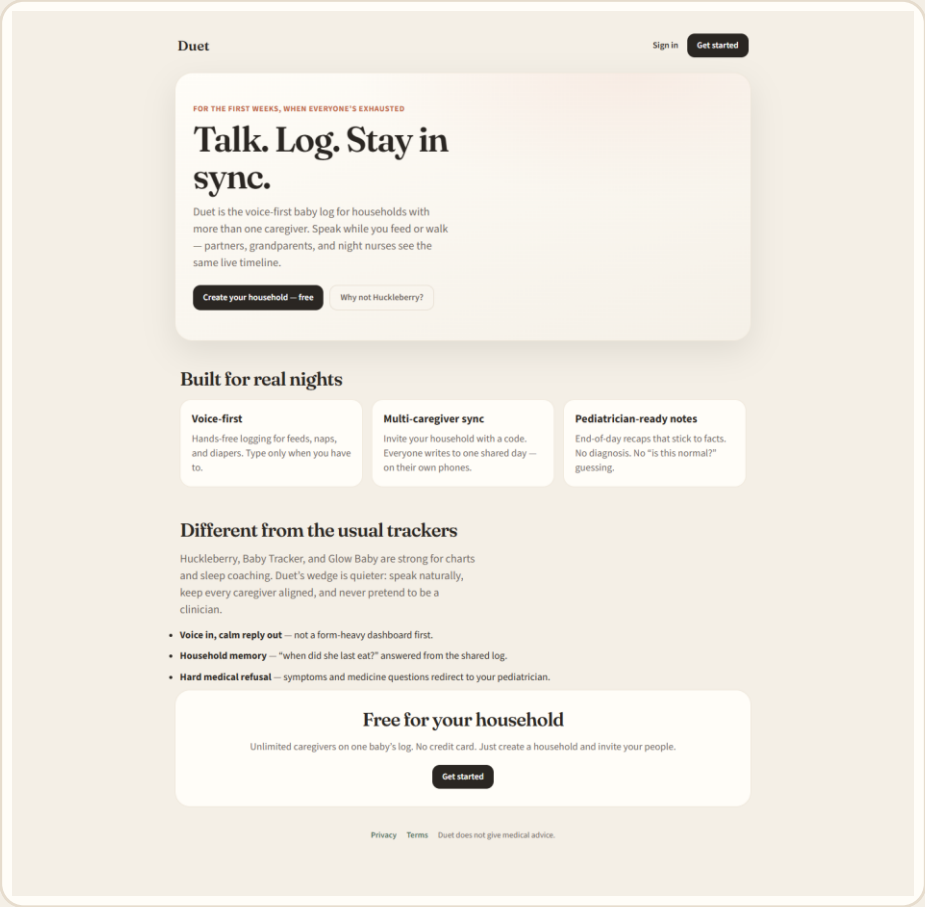
Alex on a phone logs by voice



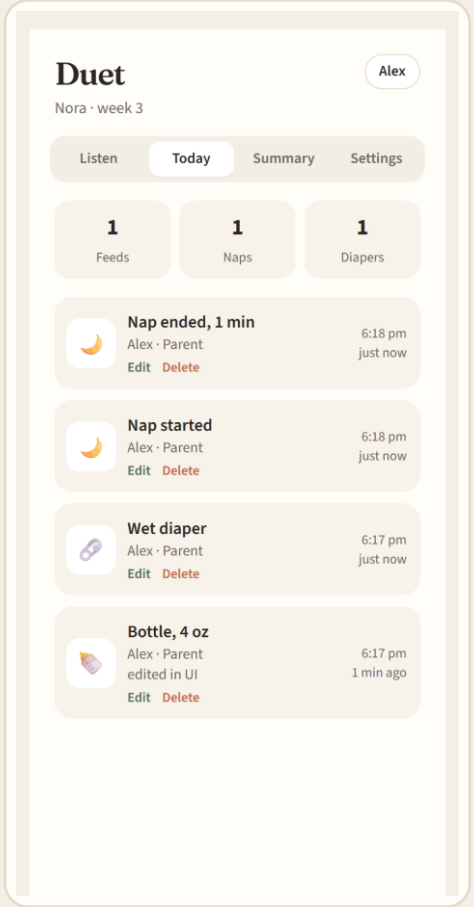
Sam on a laptop sees the shared timeline

RESPONSIVE

Works on a phone and on a full desktop screen



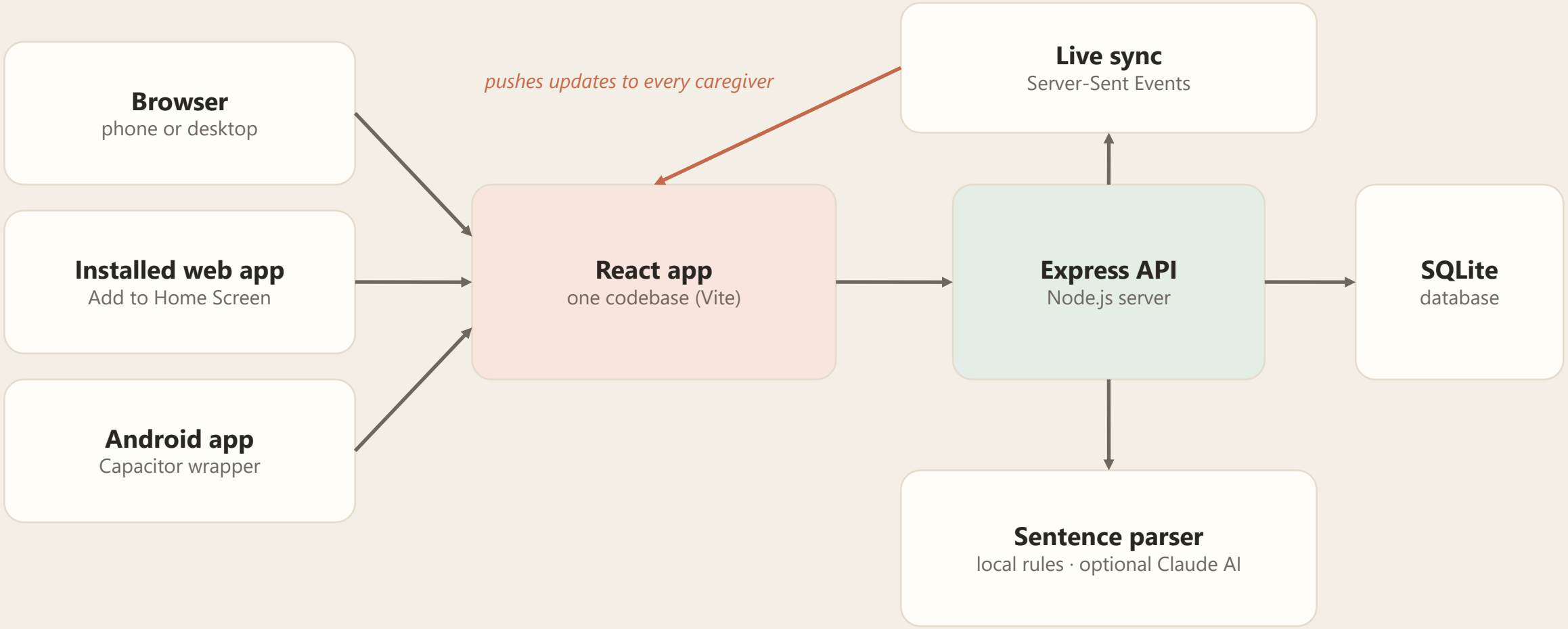
Desktop: full-width landing page



Phone: Today timeline

HOW IT IS BUILT

Architecture



ONE CODEBASE

Three ways to run the same app

Website

Opens in any browser.

Phone layout on small screens, full-width landing page on desktop.

Same accounts and live sync everywhere.

Installable web app

Web app manifest with name, colors and "standalone" display.

Icons from 48 to 512 px, plus a maskable icon.

Service worker stores the app for offline use.

Separate iPhone tags and home-screen icon.

Android app

Capacitor wraps the same build.

App ID com.duet.household, Duet icon and splash.

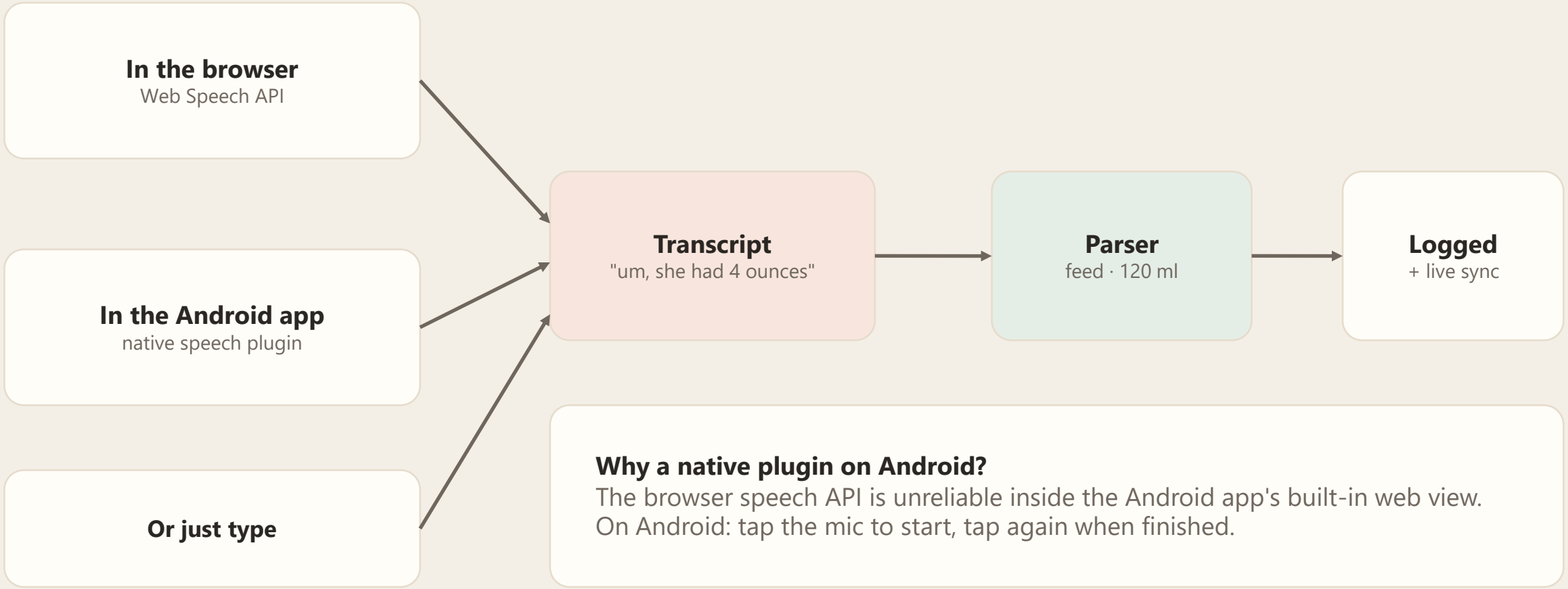
Emulator reaches the laptop at 10.0.2.2:5000.

Microphone permission for voice.

No forked code: one command (npm run cap:sync) copies the latest web build into the Android project.

VOICE

Voice works in the browser and in the Android app



TESTING

Three layers of automated tests

31 — **Browser tests**
Does the real app work on phone and desktop, live, even offline?

73 — **End-to-end tests**
Does every feature work for two caregivers, and keep outsiders out?

11 — **Unit tests**
Does Duet understand the sentence?

115
automated checks

TESTING

What the browser test actually does

1 Alex · phone

Alex signs up

2 Alex · phone

Creates household, gets invite code

3 Sam · desktop

Sam joins with the code

4 Alex · phone

Alex logs by voice

5 Sam · desktop

Sam sees it live

6 Sam · desktop

Sam edits and deletes

7 Alex · phone

Alex sees changes live

8 Alex · phone

Alex goes offline, logs, reconnects

Every step is checked automatically, including that the other person's screen updates without a refresh.

RESULTS

115 of 115 checks passed

Unit + end-to-end	Checks	Result
Sentence understanding (unit)	11	Pass
Accounts and sign-in	10	Pass
Households and invites	9	Pass
Logging by voice and text	13	Pass
Edit and delete	6	Pass
Summary, settings, export	12	Pass
Android app reaching the server	3	Pass
Installable web app files	11	Pass
Android project setup	8	Pass
Server health	1	Pass

Real browser (Edge)	Checks	Result
Layout and installability	6	Pass
Signup and joining	5	Pass
Voice and text with live sync	7	Pass
Edit and delete with live sync	3	Pass
Summary and settings screens	4	Pass
Offline behavior	4	Pass
Session and no JavaScript errors	2	Pass

115 passed · 0 failed

11 unit · 73 end-to-end · 31 browser

WHY TEST?

Testing found 4 real bugs, all fixed

Removing a caregiver crashed the server

Failed for anyone who had logged an entry.

Found by: End-to-end test
Fix: Removal now revokes access but keeps their history; they can rejoin.

"She had 4 ounces" wasn't logged

The parser knew "ounce" but not "ounces" or "drank".

Found by: End-to-end test
Fix: Added the missing words and 2 new unit tests.

Opening the app offline signed you out

The failed account check erased the saved login.

Found by: Code review for the offline test
Fix: Stays signed in, shows "You're offline", reloads on reconnect.

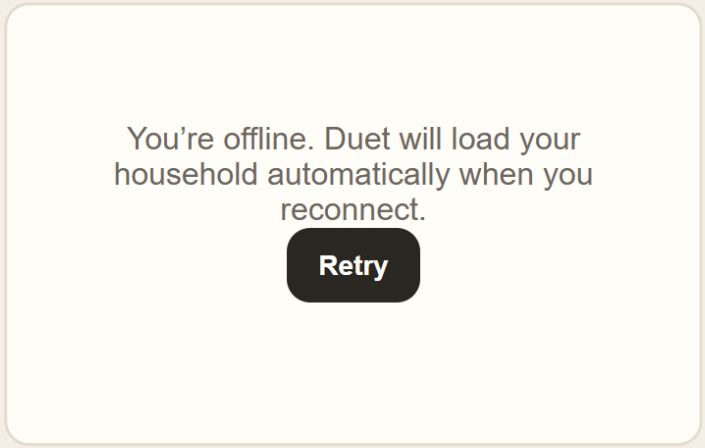
Header button had no visible label

Dark text on a dark button: a blank black bar.

Found by: Reviewing test screenshots
Fix: Fixed the color; a test now checks every main button.

OFFLINE

No signal? Duet still opens



What you see with no connection

Stored on the device

The service worker keeps the app shell, so it opens with no network.

Clear message, still signed in

"You're offline. Duet will load your household automatically when you reconnect."

Entries are queued

Anything logged offline is saved on the phone.

Syncs on reconnect

The app reloads itself and the queued entry reaches every caregiver.

HONEST LIMITS

What has not been tested yet

Not tested	Why	How to check it
Running in the Android emulator	Android Studio not installed on the test computer	Install Android Studio, run npm run cap:sync then npm run cap:open, press Run
Real spoken voice	Automated tests cannot use a microphone (a stand-in recognizer was used)	Open in Chrome, tap the mic, say "Just gave her a 3 ounce bottle"
iPhone "Add to Home Screen"	No iPhone or Mac available	Safari → Share → Add to Home Screen

Everything else, including the Android project's settings, is covered by the 115 automated checks.

TRY IT

Run it and test it yourself

```
# start the app
npm start

# run every test
npm test
npm run test:e2e
npm run test:ui

# Android
npm run cap:sync
npm run cap:open
```

Open the app

http://localhost:5000 in any browser.
On a phone: install it from the browser menu or Share sheet.

Read the full report

docs/TEST_REPORT.md lists every check, the bugs found, and the raw test output.



Thank you

Questions? Happy to do a live demo.

115 automated checks · 4 bugs found and fixed · one codebase for web and Android